ABHINANDAN BHATIA

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EDUCATION

Shiv Nadar Institution of Eminence ..... Aug 2024

B.S Math (Physics & Economics Minor)

CGPA: 8.55/10 | Dean's List Awardee (Monsoon 2024)

SELECTED ACHIEVEMENTS

- **Selected Delegate**, **Harvard Project for Asian and International Relations (HPAIR)**, Harvard University (2026)
- **Rank 1 / 3,461** — Guru Nanak Dev University Undergraduate Entrance Test
- **Research Intern, University of Delhi** — Theoretical & Computational Astrophysics
- **Selected Participant, IIIT Delhi Undergraduate Summer Program** in Mathematics (~10% acceptance)
- **Published Contributor— Indian Express & Hindustan Times** (National Editions)

RESEARCH EXPERIENCE

Incoming Research Intern — University of California, Santa Cruz (UCSC), Department of Economics

Summer 2026 (Incoming) | Supervisor: Prof. Grace Gu

- Selected to contribute to research on wage stagnation and long-term labor market dynamics
- Curating and analyzing economic datasets; conducting literature review and empirical analysis
- Applying statistical and econometric methods to study wage trends and income dynamics
- Investigating structural drivers of wage growth, productivity, and inequality

IIIT Delhi Undergraduate Summer Program in Mathematics

June 2025

- Selected for competitive national program (~10% acceptance rate)
- Advanced coursework in abstract algebra, real analysis, and geometry
- Participated in proof-intensive seminars and faculty research lectures

Research Intern — D.S. Kothari Centre, Miranda House, University of Delhi

June–July 2025 | Supervisor: Dr. Richa Kundu

- Analyzed Gaia EDR3 astrometric and photometric data of globular clusters; implemented Gaussian Mixture Models and Bayesian membership classification to distinguish cluster and field populations
- Constructed extinction-corrected color–magnitude diagrams and fitted stellar isochrones to study evolutionary structure and global cluster properties
- Investigated the spatial distribution of extra-tidal stars and contributed to drafting the final research manuscript

SELECTED CONFERENCES & ACADEMIC EVENTS

Harvard Project for Asian and International Relations (HPAIR) ..... Feb 2026

Harvard University, Cambridge, MA, USA | February 2026

- Selected as a student delegate for a competitive international conference focused on global policy, economics, and international relations

Poster Presentation — Ashoka Physics Student Symposium ..... Sept 2026

Ashoka University, India | September 2025

- Presented research poster on Gaia-based membership modeling and extra-tidal structure analysis of globular cluster NGC 7078 (M15)
- Discussed probabilistic modeling techniques and astrophysical implications with faculty and student researchers

Winter School in Economics — Delhi School of Economics (DSE) ..... Dec 2024

Delhi School of Economics, India | December 2024

- Attended plenary lectures by leading economists including Rohini Pande, Samuel Kortum, Arvind Panagariya, Julia Cagé, and Thomas Piketty
- Participated in academic sessions covering microeconomics, international economics, political economy, and mathematical economics

ACADEMIC & PUBLIC ENGAGEMENT

Published Contributor — Indian Express & Hindustan Times (2022–) .....

Selected published commentaries include:

- *Empathy Needed* — Children's mental health in post-pandemic schooling (*Indian Express*, 22/02/2022)
- *Rule Bound* — Academic flexibility in university policy (*Indian Express*, 25/04/2022)
- *Need for Expertise* — Role of scientific and policy experts in governance (*Indian Express*, 07/11/2025)
- *India's Opportunity* — Strengthening global research ecosystems (*Hindustan Times*, 04/06/2025)
- *GST Cuts* — Economic policy and domestic stability (*Hindustan Times*, 26/09/2025)

Invited Talk — "Math, Logic, and Language" .....

Department of English, Shiv Nadar Institution of Eminence | Monsoon 2025

- Delivered invited lecture to English majors (ENG451 Screenwriting) at the invitation of Prof. Dr. Sam North (University of Exeter, UK)
- Explored connections between Russell, Wittgenstein, and Chomsky; discussed formal language theory and parallels between mathematics, programming, and natural language
- Lecture slides incorporated into official course resources

Inaugural Lecture — The Mathematics Society, SNIoE .....

Game Theory: Winning Mathematically" | Monsoon 2025

- Introduced historical foundations of game theory and modeling of strategic interaction
- Covered non-cooperative games, Prisoner's Dilemma, payoff matrices, and Nash equilibrium

SKILLS

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Communication & Representation: Academic writing and research presentation; published contributions in national newspapers; experience presenting interdisciplinary material to diverse academic audiences; engagement in cross-disciplinary academic dialogue.

Technical Skills: Quantitative and probabilistic modeling; statistical analysis and data-driven research; Python-based scientific computing; LaTeX documentation.

COURSEWORK

Selected Coursework

Real Analysis I–II, Group Theory, Rings & Fields, Differential Geometry, Probability, Quantum Mechanics, Theoretical Mechanics, Electrodynamics. Principles of Macroeconomics, Transformational Leadership.